ANDERSON JUNIOR COLLEGE

2013 JC2 PRELIMINARY EXAMINATIONS

CHEMISTRY **8872/01**

Higher 1 **23 September 2013**

Paper 1 Multiple Choice **50 minutes**

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **17** printed pages.

Section A

For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 *Use of the Data Booklet is relevant to this question.*

Which of the following statements is **incorrect**?

A 35.5 g of chlorine gas contains 6.0×10^{23} chlorine atoms.

B 24 dm^3 of hydrogen gas at 25 °C and 1 atm contains 1.2×10^{24} hydrogen atoms.

C 500 cm^3 of 1 mol dm^{-3} aqueous magnesium nitrate(V) contains 3.0×10^{23} nitrate(V) ions.

D 4 g of helium gas contains 6.0×10^{23} helium atoms.

2 Which mass of gas would occupy a volume of 3 dm^3 at 25 °C and 1 atmospheric pressure?
[1 mol of gas occupies 24 dm^3 at 25 °C and 1 atmospheric pressure]

A 3.2 g O_2 gas

B 5.6 g N_2 gas

C 8.0 g SO_2 gas

D 11.0 g CO_2 gas

3 The table below contains incomplete information about the three isoelectronic ions **X**, **Y** and **Z**. The atoms of **Y** and **Z** are isotopes.

ion	mass number	atomic number	number of neutrons	number of electrons	charge
X	55	25			***b***
Y	56	26			
Z	***a***		31		+3

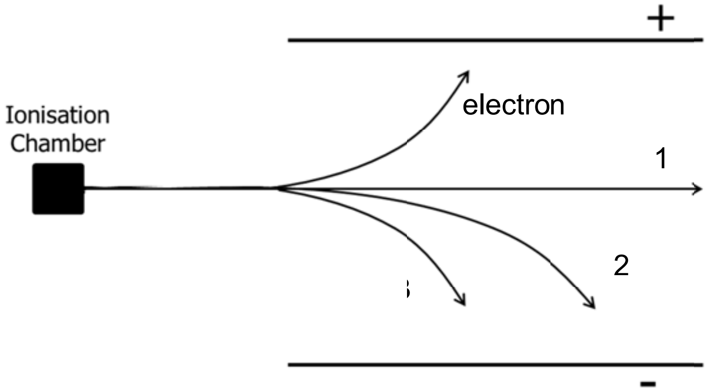
What are the values of ***a*** and ***b***?

	a	***b***
A	56	+2
B	56	+3
C	57	+2
D	57	+3

4 *Use of the Data Booklet is relevant to this question.*

$^{243}_{94}$Pu can undergo a natural radioactive decay, where one of its electrons enters the nucleus to change a proton into a neutron, to form a new element **M**.

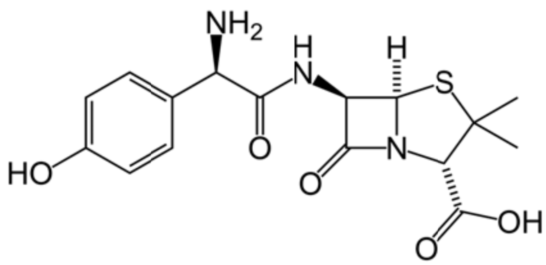
When **M** is put in an ionisation chamber, it emits a high energy α-particle (which is a ^{4}He nucleus).



What is the identity of the element **M** and the path of the emitted α-particle in an electric field?

	Chemical symbol of **M**	Path of (α–particles)
A	$^{243}_{93}$**M**	2
B	$^{243}_{95}$**M**	1
C	$^{244}_{93}$**M**	2
D	$^{244}_{95}$**M**	3

5 *Amoxicillin* is one of the most common antibiotics prescribed for children.



Amoxicillin

How many lone pairs of electrons are present in this molecule?

A 13 **B** 15 **C** 17 **D** 19

6 The African weaver ant defends its territory by spraying an intruder with a mixture of compounds. The ease by which these compounds are detected by other ants depends upon the volatility, which decreases as the strength of the intermolecular forces in the compound increases.

Which compound in the mixture would be the most volatile?

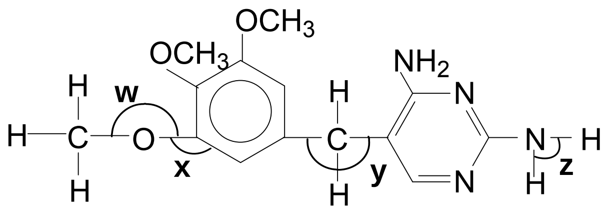
A $CH_3CH_2CH_2CH_2CH_2CH_3$

B $CH_3CH_2CH_2CH_2CH_2CHO$

C $CH_3CH_2CH_2CH_2CH_2COOH$

D $CH_3CH_2CH_2CH_2CH_2OH$

7 *Trimethoprim* (TMP) is used for the treatment and prevention of urinary tract infection, traveller's diarrhea, respiratory and middle ear infections. It has the following structure.



In which sequence is the bond angles quoted in decreasing order?

A **w = y > x > z**

B **x > y > w > z**

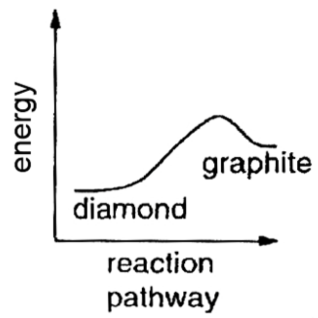
C **x > y > z > w**

D **w > x > y > z**

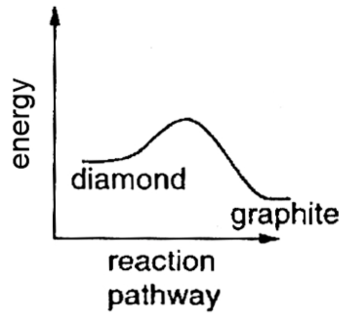
8 The conversion of diamond into graphite is an exothermic reaction. Diamond does not readily change into graphite.

Which reaction pathway correctly represents this conversion?

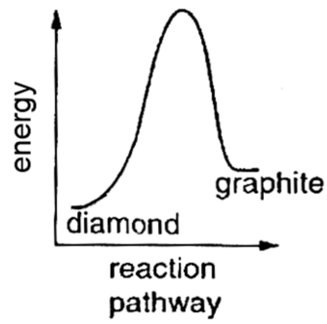
A



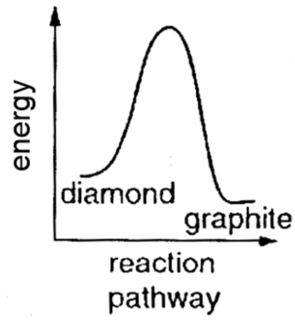
B



C



D



9 The table shows the enthalpy change of neutralisation, Δ*H*, for the various acids and bases listed.

acid	base	Δ*H* / kJ mol^{-1}
sulfuric acid	sodium hydroxide	–57.0
P	sodium hydroxide	Less exothermic than –57.0
sulfuric acid	**Q**	Less exothermic than –57.0
R	potassium hydroxide	–57.0

What are the likely identities of **P**, **Q** and **R**?

	P	Q	R
A	hydrochloric acid	lithium hydroxide	nitric acid
B	phosphoric acid	ammonia	ethanoic acid
C	nitric acid	lithium hydroxide	hydrochloric acid
D	hydrogen cyanide	ammonia	nitric acid

10 Ammonia is manufactured by the Haber process, in an exothermic reaction.

Assuming that the amount of catalyst remains constant, which change will **not** bring about an increase in the rate of the forward reaction?

A decreasing the size of the catalyst pieces

B increasing the pressure

C increasing the temperature

D removing the ammonia as it is formed

11 The value of the equilibrium constant, K_c, for the reaction to form ethyl ethanoate from ethanol and ethanoic acid is 4.0 at 60 °C.

$$C_2H_5OH + CH_3CO_2H \rightleftharpoons CH_3CO_2C_2H_5 + H_2O$$

What is the number of moles of ethyl ethanoate formed when 1.0 mol of ethanol and 1.0 mol of ethanoic acid are allowed to reach equilibrium at 60 °C?

A $\frac{1}{3}$ **B** $\frac{2}{3}$ **C** $\frac{1}{4}$ **D** $\frac{3}{4}$

12 Bromocresol green is an acid–base indicator with a pH range of 3.8 to 5.4. The acidic colour of the indicator is yellow and the alkaline colour is blue.

Two drops of the indicator are added to each of the four aqueous solutions listed below.

Which solution has its colour correctly stated?

	Solution	**Colour**
A	Blood plasma	Green
B	Aqueous solution of $MgCl_2$	Yellow
C	Dilute HCl of concentration 3.0×10^{-5} mol dm^{-3}	Yellow
D	Aluminium oxide added to water	Blue

13 Which of the following equations does **not** show a Brønsted acid−base reaction?

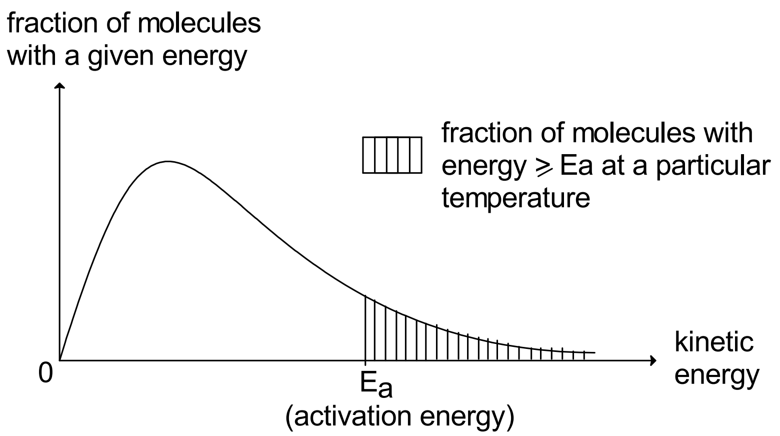
A $Na^+NH_2^- + NH_4^+Cl^- \longrightarrow Na^+Cl^- + 2NH_3$

B $2NCl_3 + 6NaOH \longrightarrow N_2 + 3NaCl + 3NaOCl + 3H_2O$

C $OCl^- + H_2O \longrightarrow OH^- + HOCl$

D $H_2O + SO_3^{2-} \longrightarrow OH^- + HSO_3^-$

14 Below shows the Maxwell–Boltzmann distribution curve for a chemical reaction in living systems.



Which of the following deductions is correct?

A The total area under the graph will increases at high temperature.

B The shaded area under the graph will decrease when enzymes are denatured.

C An increase in the surface area of the reactants decreases the E_a.

D The order of reaction with respect to the chemical reaction is first order when the kinetic energy is low.

15 Why is the ionic radius of a chloride ion larger than the ionic radius of a sodium ion?

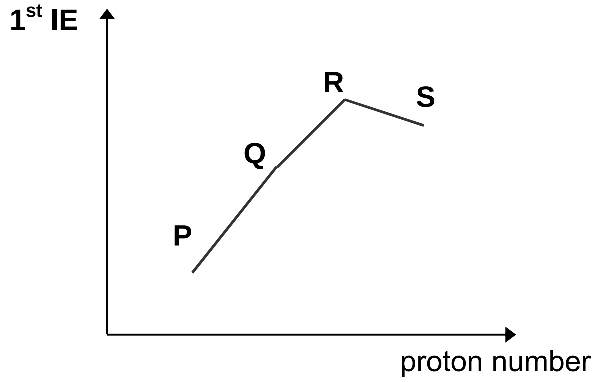
A A chloride ion has one more occupied electron shell than a sodium ion.

B Chlorine has a higher proton number than sodium.

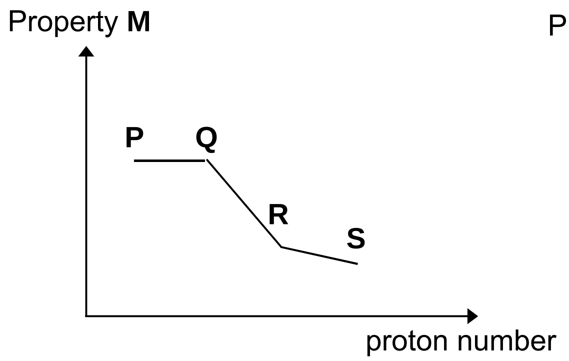
C Ionic radius increases regularly across the third period.

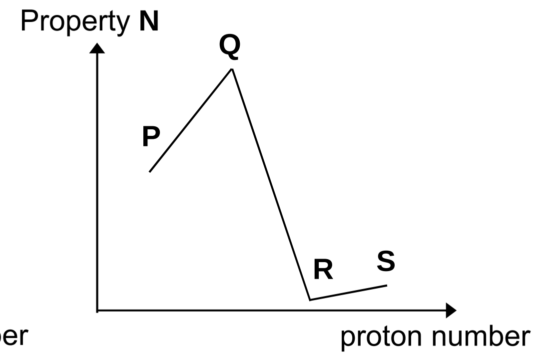
D Sodium is a metal whilst chlorine is a non−metal.

16 The diagram represents the first ionisation energy of four consecutive elements in the third period of the Periodic Table, with element **R** showing two oxidation states in its chlorides.



The sketches below represent another two properties of these elements.





What are properties **M** and **N**?

	property **M**	property **N**
A	pH of oxides in water	melting point
B	ionic charge	5^{th} ionisation energy
C	ionic radius	melting point
D	electrical conductivity	5^{th} ionisation energy

17 Elements **X** and **Y** are both in Period 3 of the Periodic Table.

When the chloride of **X** is added to water, it reacts and a solution of pH 1 is produced.

When the chloride of **Y** is added to water, it dissolves and a solution of pH 7 is produced.

Which statement explains these observations?

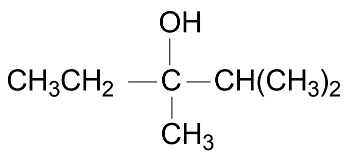
A Both chloride hydrolyse in water.

B **X** is phosphorus and **Y** is aluminium.

C **X** is silicon and **Y** is sodium.

D **X** is sodium and **Y** is phosphorus.

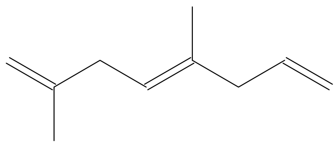
18 How many possible products (including stereoisomers) can be formed when the following compound is reacted with excess concentrated sulfuric acid at 180 °C?

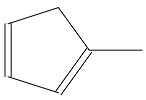


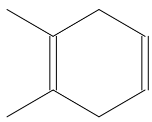
A 3 **B** 4 **C** 5 **D** 6

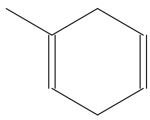
19 A hydrocarbon, on heating with an excess of hot concentrated acidified $KMnO_4$(aq), produces CH_3COCH_2COOH as the only organic product.

Which of the following is **not** a possible structure of the hydrocarbon?

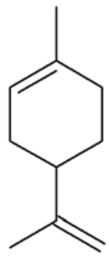
A 

B 

C 

D 

20 Limonene is a constituent of lemon oil and is a useful starting material for the manufacture of perfumes.

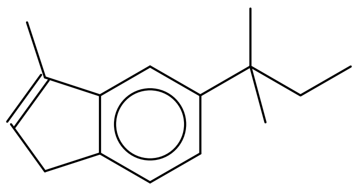


limonene

How many σ and π bonds are there in a molecule of limonene?

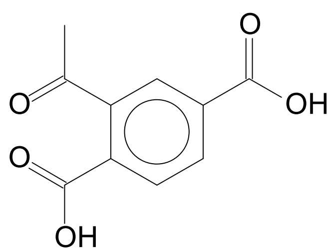
	σ	π
A	8	4
B	10	2
C	24	4
D	26	2

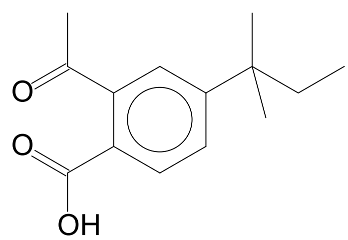
21 Compound **Z**, which has an aromatic ring structure, is subjected to oxidative degradation under suitable conditions.

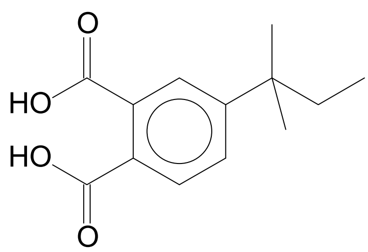


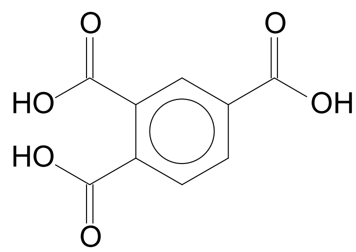
Z

What are the most likely organic products from this reaction?

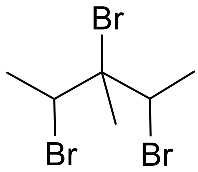
A 

B 

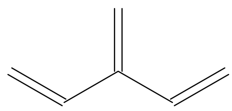
C 

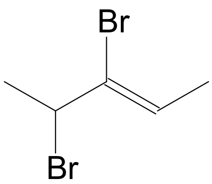
D 

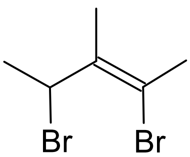
22 The following compound is treated with hot ethanolic potassium hydroxide.

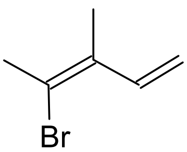


Which of the following is **not** a possible product?

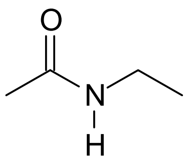
A 

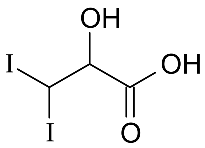
B 

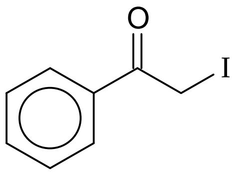
C 

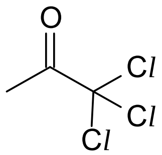
D 

23 Which of the following will **not** give tri–iodomethane on warming with alkaline aqueous iodine?

A 

B 

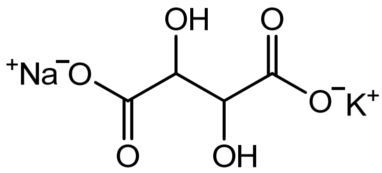
C 

D 

24 Which of the following involves a pair of addition reactions?

	Reaction 1	**Reaction 2**
A	But–1–ene → 2–chlorobutane	Butanone + 2,4–dinitrophenylhydrazine
B	Butanal → 2–hydroxypentanoic acid	But–1–ene + PCl_5
C	But–1–ene → butanone	But–1–ene + HCN
D	But–1–ene → butan–2–ol	Butanal + HCN

25 Potassium sodium tartrate, also known as Rochelle salt, is used medicinally as a laxative and has the following structure.



Which of the following could be part of the reaction sequence to synthesise Rochelle salt?

A $CH_2BrCH_2Br \xrightarrow[\text{reflux}]{\text{alcoholic KCN}} \text{intermediate} \xrightarrow[\text{reflux}]{\text{KOH(aq) + NaOH}} \text{Rochelle salt}$

B $HOCH_2(CH_2)_2Cl \xrightarrow[\text{reflux}]{\text{alcoholic KCN}} \text{intermediate} \xrightarrow[\text{reflux}]{KMnO_4\text{ , KOH}} \text{Rochelle salt}$

C $CHOCHO \xrightarrow[10^\circ C\text{ - }20^\circ C]{\text{HCN(aq) + NaCN}} \text{intermediate} \xrightarrow[\text{reflux}]{\text{KOH(aq) + NaOH}} \text{Rochelle salt}$

D $CH_3COCH_3 \xrightarrow[10^\circ C\text{ - }20^\circ C]{\text{HCN(aq) + NaCN}} \text{intermediate} \xrightarrow[\text{reflux}]{KMnO_4\text{ , NaOH}} \text{Rochelle salt}$

Section B

For each of the question in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

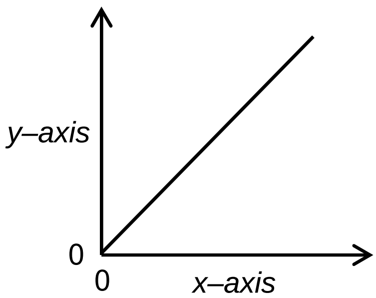
A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

26 The kinetics of a first order reaction **X** $\longrightarrow$ **Y** was investigated under different conditions.

The table shows pairs of quantities that were plotted as graphs.

Which pairs gave the following graph?



	y–axis	*x–axis*
1	Rate	[**X**]
2	[**Y**]	time
3	[**X**]	[**Y**]

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

27 **P** reacts with **Q** according to the equation below.

$$\mathbf{P}(g) + \mathbf{Q}(g) \rightleftharpoons \mathbf{R}(g) \qquad \Delta H < 0$$

Which statement about the reaction is correct?

1 When the activation energy of the forward reaction is decreased, the rate constant of the reverse reaction increases.

2 Increasing temperature will lower the activation energy, resulting in a greater fraction of **R** molecules with energy greater than activation energy.

3 The activation energy of the forward reaction is larger than the activation energy of the backward reaction.

28 Which of the following correctly describes the compounds formed when Period 3 elements from Na to A*l* react with oxygen?

1 covalent character increases

2 solubility in water decreases

3 pH of solution when dissolved in water decreases

29 In the European Union, petrol is often blended with ethanol. Which reagent (s) could be used to detect the presence of ethanol in the petrol which also consist of a mixture of alkanes and alkenes?

1 Na

2 $KMnO_4$

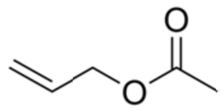
3 2,4−dinitrophenylhydrazine

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

30 Allyl acetate is an organic compound with following skeletal formula.



Which of the following statements are true about allyl acetate?

1 It gives an orange precipitate when heated with Brady's reagent.

2 It has three sp^2 carbon atoms and two sp^3 carbon atoms.

3 It can be formed from reacting ethanoic acid and propan–1,3–diol, $HOCH_2CH_2CH_2OH$ in concentrated sulfuric acid and heating under reflux.

H1 Chemistry 8872 **AJC 2013 JC2 PRELIM** **Paper 1 – 30 marks**

Question Number	*Key*	*Question Number*	*Key*
1	**C**	16	**A**
2	**C**	17	**C**
3	**C**	18	**B**
4	**A**	19	**D**
5	**B**	20	**D**
6	**A**	21	**C**
7	**C**	22	**B**
8	**D**	23	**A**
9	**D**	24	**D**
10	**D**	25	**C**
11	**B**	26	**D**
12	**D**	27	**D**
13	**B**	28	**A**
14	**B**	29	**D**
15	**A**	30	**C**